Lesson for the course “Human-Computer Interaction”

Sensors, actuators, and types of interaction

Prof. Luca Turchet

UNIVERSITY OF TRENTO - Italy

Project: an app in conjunction with a smart object or smart environment

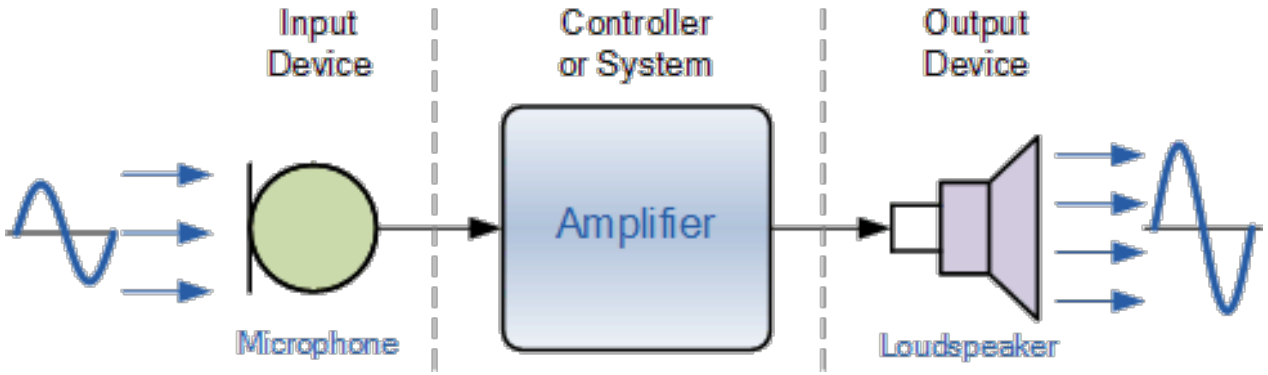
- For your exam project you can create an app that interfaces with a smart object or a smart environment
- App: e.g., app for mobiles (smartphone, tablet) or a program for a laptop/desktop computer

Project: an app in conjunction with a smart object or smart environment

- Smart objects are objects that **embed electronics**, including sensors, actuators, wireless connectivity, and a processing unit
- "Smart objects": they have intelligence (e.g., AI systems, or in general ability to analyze data and exchange data with other devices)
- **What are the services you offer to users?**
  - These services need to be derived from user research

Sensors and actuators

- **Sensors** are devices that **detect** a physical change in some characteristic that change in response to some excitation and **convert** that into an electrical signal
  - E.g., a microphone converts sound waves into electrical signals
- **Actuators** are devices that **convert** an electrical signal into a physical quantity that **affects** the world
  - E.g., a loudspeaker converts electrical signals (e.g., produced by an amplifier) into sound waves



Augmenting objects with sensors

- Ordinary objects can be enhanced with input and output facilities so that they perform their original function as well as they provide a way for **data to be collected about the user's behaviour**
- The data from the device may be uploaded to the user's online dashboard (e.g., via Bluetooth to a tablet) and the information presented to the user is also available on the mobile devices
- Statistics are displayed and recommendations are provided based on real-time data
- **Combining data** from several objects (belonging to different users) can result in new knowledge available to the user **to make suggestions for lifestyle improvements** (through Big Data analysis techniques)

Example of app in conjunction with a smart object or smart environment

- Example of an **existing product**: an app for a smart washing machine
- E.g., data of the user interaction is captured and transmitted to a server, which analyzes this data and recommends some services to the user



- Which services?
- Which users?

Example of app in conjunction with a smart object or smart environment

- Example of an **imaginary product**: a knife with embedded sensors that detects the pressure of the hand on the handle and the acceleration while cutting some vegetables and sends data to a mobile device about the user interaction; the app provides some kind of services to the user



- Which services?
- Which users?

Example: hapifork



- Video: https://youtu.be/-o-FyQLR8QQ

Example: touch input on clothes and other accessories



Figure 2. Touch sensors in clothing and accessories. We put (a) flexibly soldered sets of touch sensors (QProx QT110) into several device prototypes, including a (b) phone bag, (c) bicycle helmet, (d) and piece of clothing with different designs for touch areas.

Example: embedded computing for sports and entertainment

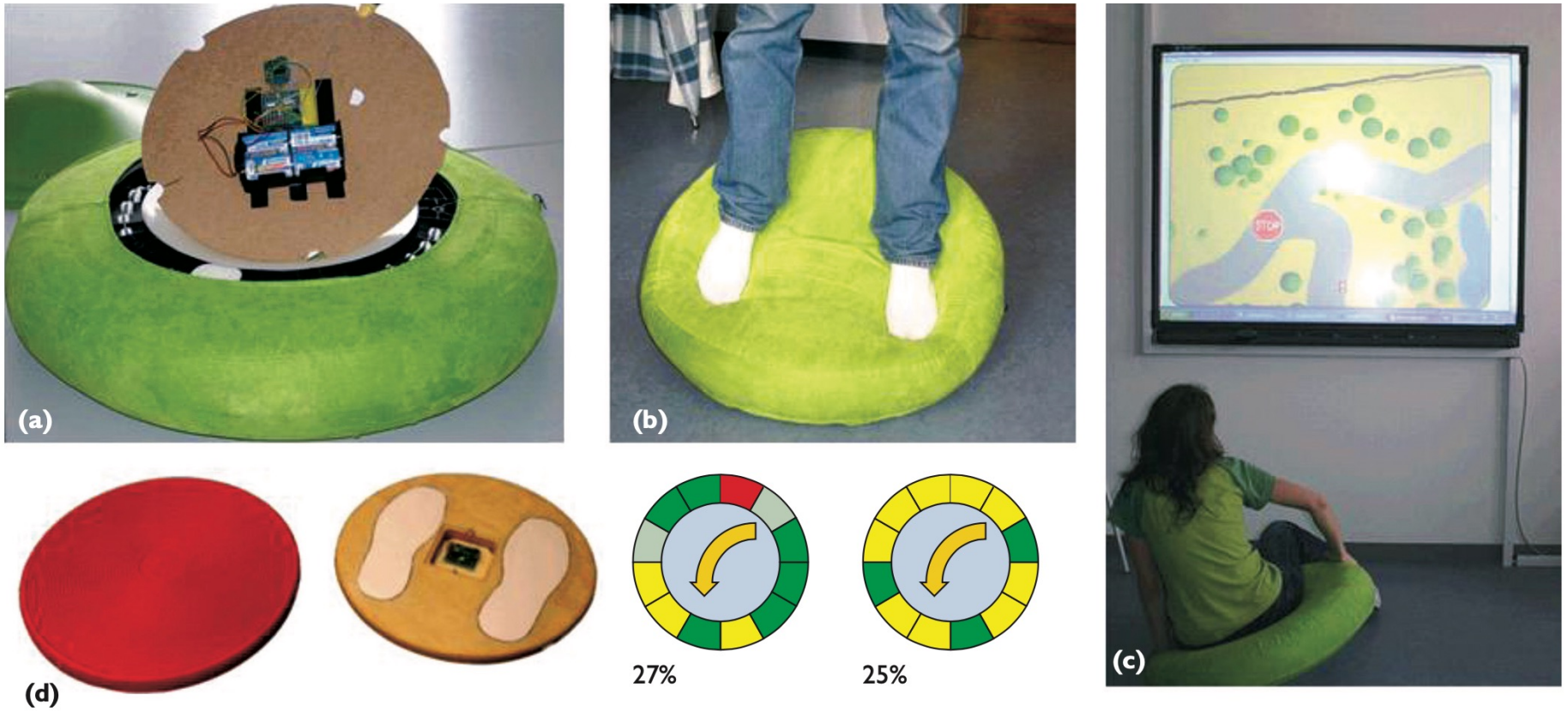


Figure 3. Embedded computing for sports and entertainment. We prototyped several projects, including (a) an Ikea cushion with built-in orientation sensors. A person can (b) stand or sit on it to control applications such as (c) an educational racing game. (d) The same technical setup enables an application used for physiotherapy and sports.

Example: enhancing environments and objects

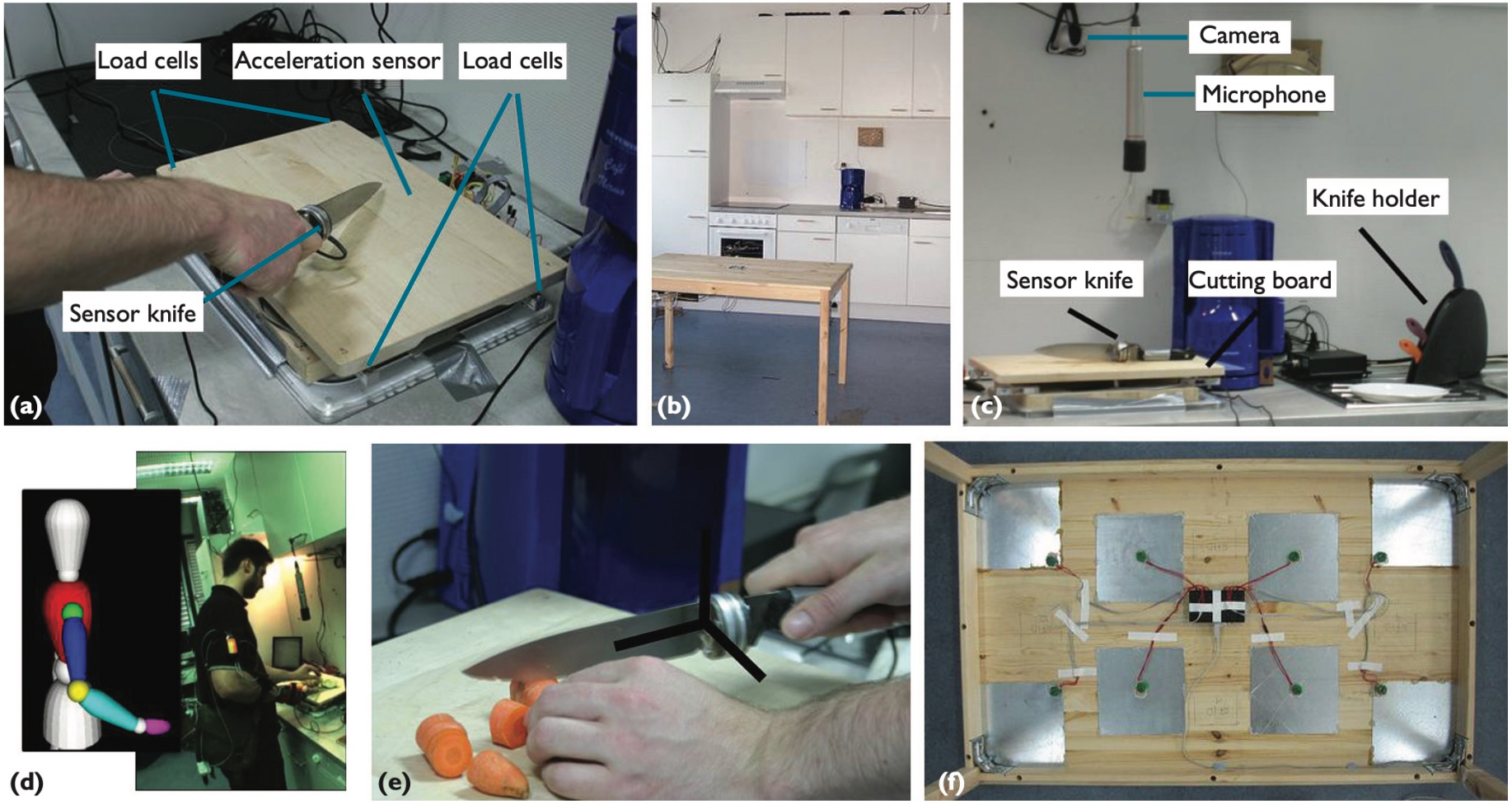


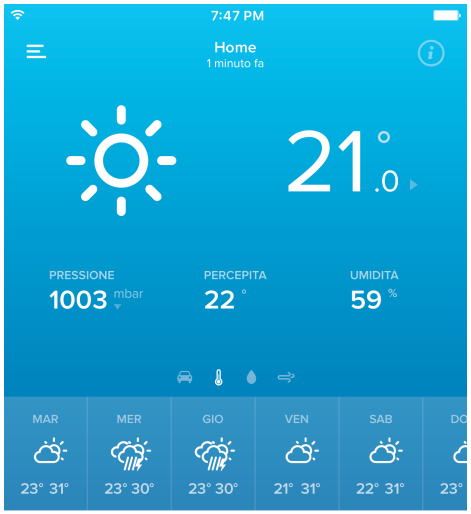
Figure 1. Netget ecology in a kitchen environment. Device prototypes include (a) an instrumented cutting board acting as a mouse pad and scale; (b) the kitchen; (c) vision and sound-based activity detection; (d) activity recognition using body-worn sensors; (e) a sensor-augmented knife netget; and (f) a table with a capacitive sensing system for recognizing table-top interactions during meals.

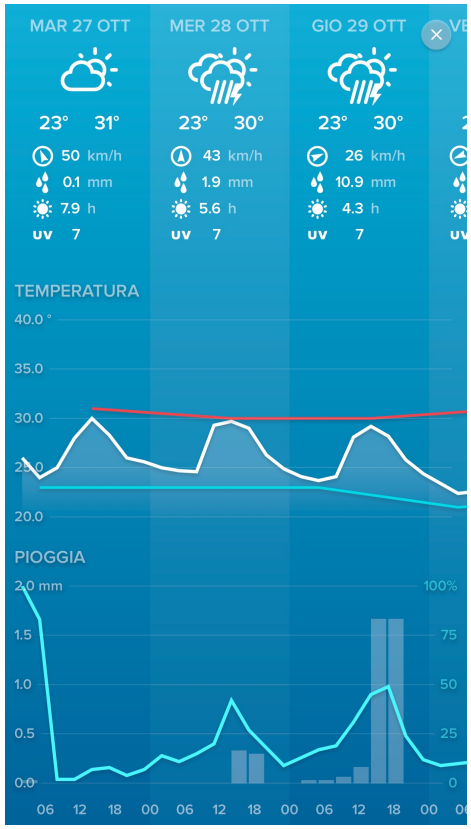
Example: NETATMO weather station

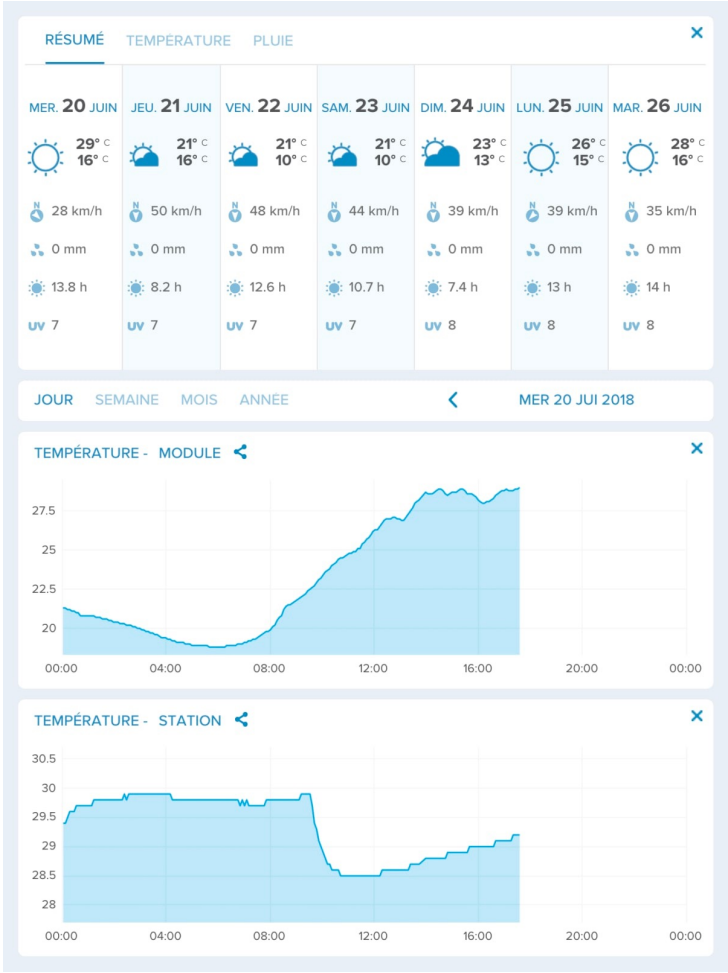
- https://www.netatmo.com/en-us/weather/weatherstation



Example: NETATMO weather station







- Check indoor environment
- Check outdoor environment
- Check past data

Visualization of collected data

- Information visualization is *"the **art and science** of representing abstract information in a visual form that enables human users to gain insight through their perceptual and cognitive capabilities"* (Robertson et al. 2009)
- How do you visualize data from the sensors to convey information to users?
- **How best to present the data collected from the real world and make it meaningful to humans?**
- Note: information about data can be conveyed also via sound and haptic sensations

System inputs: the information captured by sensors

- Body movement (e.g., via accelerometers, gyroscopes)
- Speech (via speech recognition systems)
- Facial recognition (e.g., via cameras)
- Gaze (via eye-tracker)
- Digital pen
- Biometric data (e.g., heart rate, skin conductance)
- Brain activity (e.g., via brain-computer interfaces)

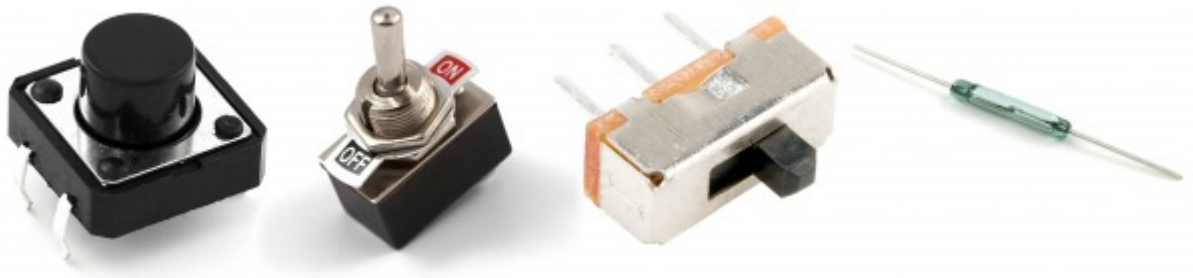
System outputs: the information conveyed by actuators

- **Visual**
  - Text, graphics, animations
  - Virtual/augmented reality
- **Auditory**
  - Speech synthesis (conversational agent)
  - Non-speech sounds (musical or non-musical sounds)
- **Haptic**
  - Force feedback
  - Vibrotactile stimulation
  - Thermal sensations
- **Scent**
  - E.g., interfaces for mobile phones
- **Taste**
  - Recent line of research

Sensors

- Many different types
- Sensors can track **gestures** of a user (e.g., hand gestures, etc.)
  - Touch-based sensor
  - Touchless sensor
- Sensors that can track **biometric parameters** of a user (e.g., heart beat, galvanic skin response, muscle tension, electroencephalogram)
- Sensors can track **properties of an environment** (e.g., light, humidity, sounds, heat)

Switch



- E.g., 2 states: on/off
- E.g., a selector with more than 2 states

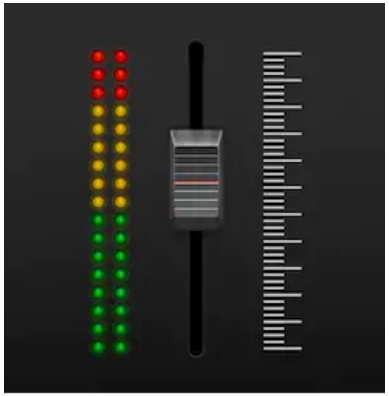
Potentiometer

- E.g., knob



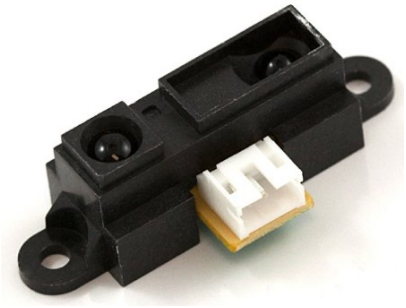


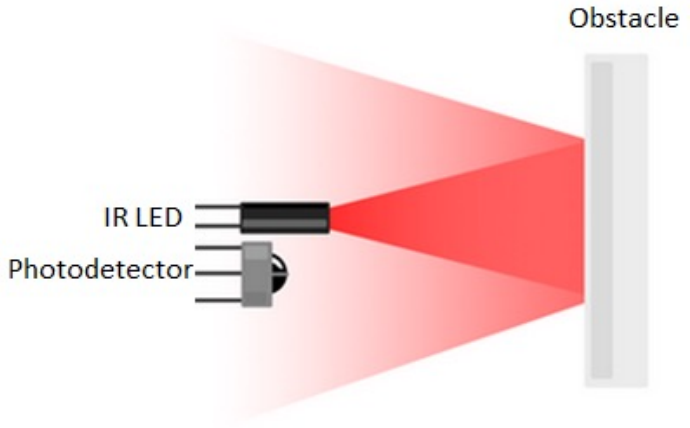
- E.g., fader



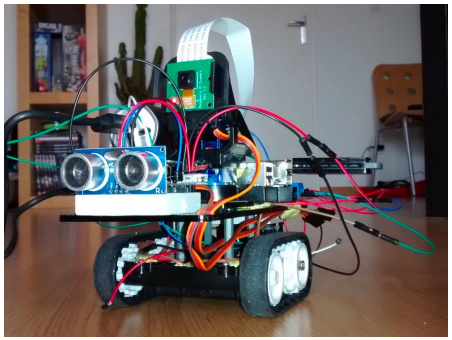


Proximity sensor





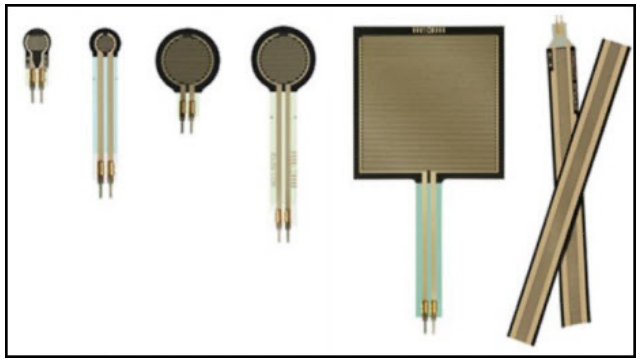
- E.g., Track the distance from obstacles in robots

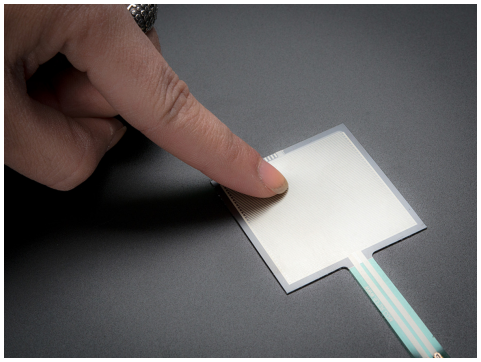


- E.g., Track the hand distance from a part of a musical instrument

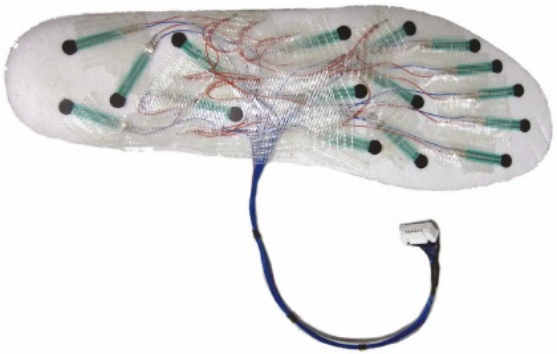


Pressure sensors





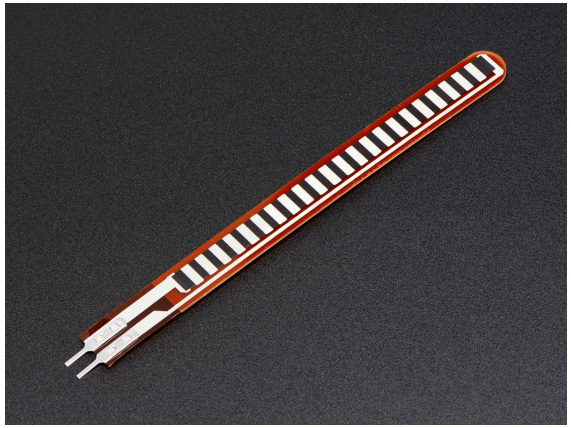
- E.g., Track the foot pressure while walking or standing

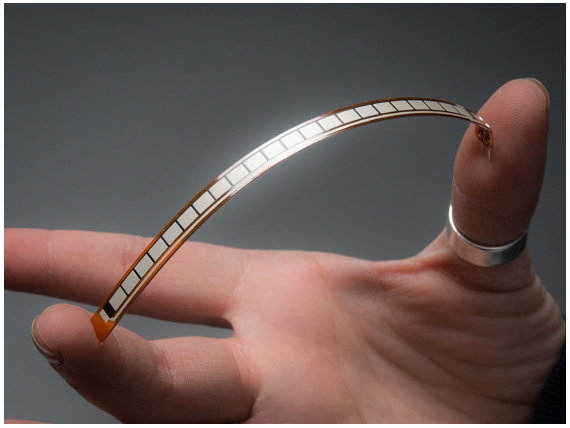


- E.g., Track finger pressure in the interface of musical instruments

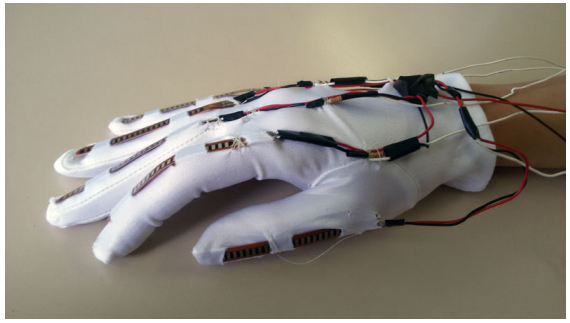


Flex sensor



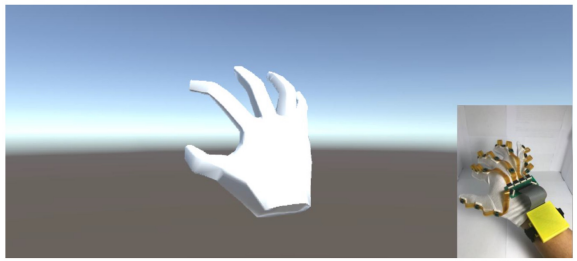


- E.g., Track fingers flexion in gloves (and then gesture recognition)

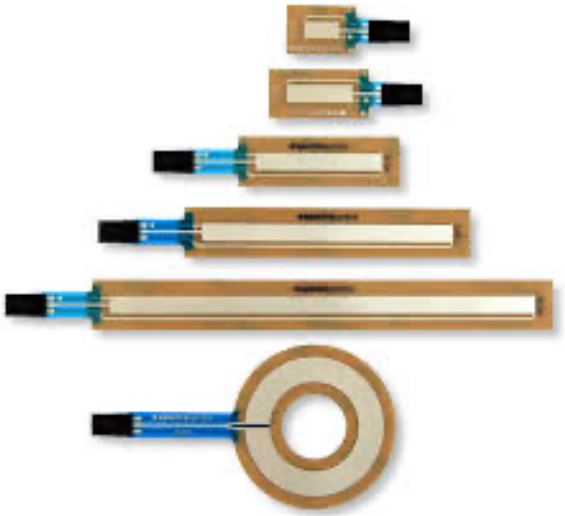




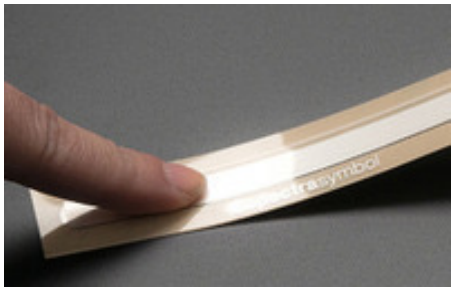


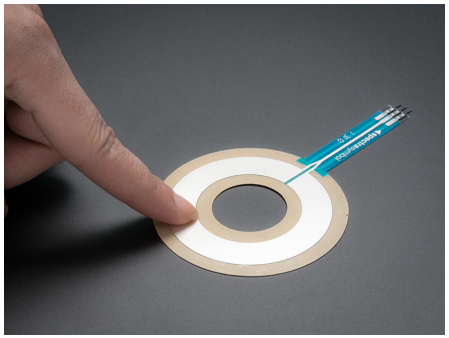


Ribbon sensors





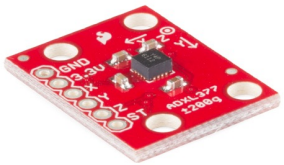


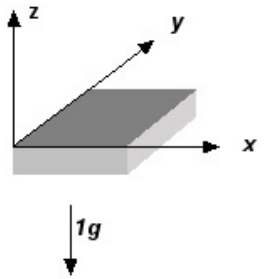


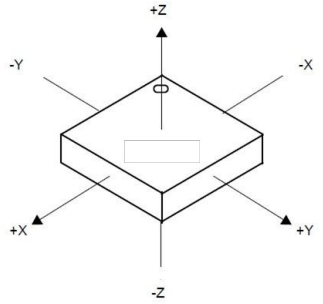
- E.g., Track fingers position on a certain area of a musical instrument



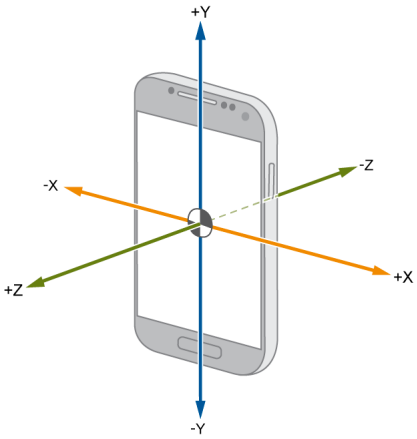
Accelerometer

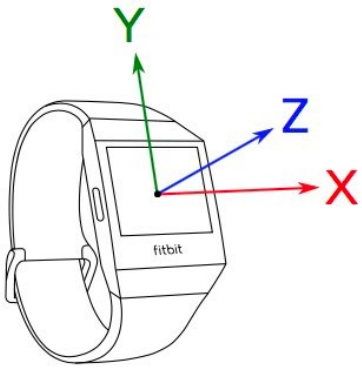






- Several applications involving the measurement of the change of velocity and repurposing this information
  - E.g., one can measure the acceleration resulting from motion, shock, or vibration
  - E.g., accelerometer embedded in a ping pong paddle



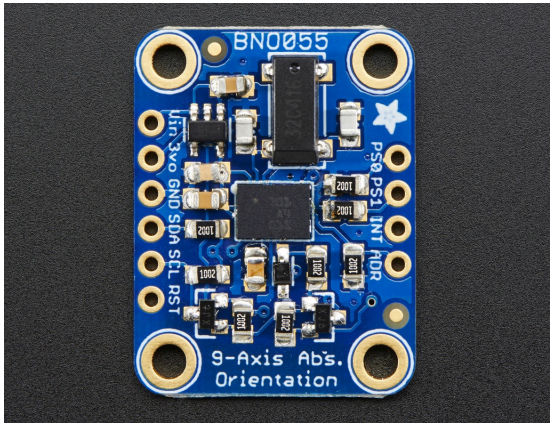


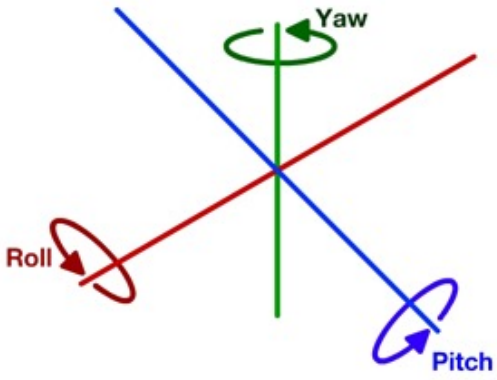


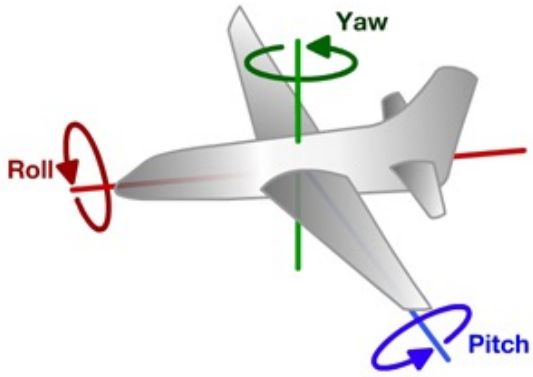


Inertial measurement unit (IMU)

- IMU: accelerometer, magnetometer and gyroscope
- Sensor fusion to give the absolute orientation







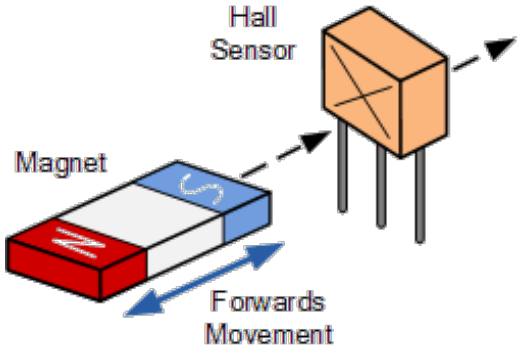
Joystick



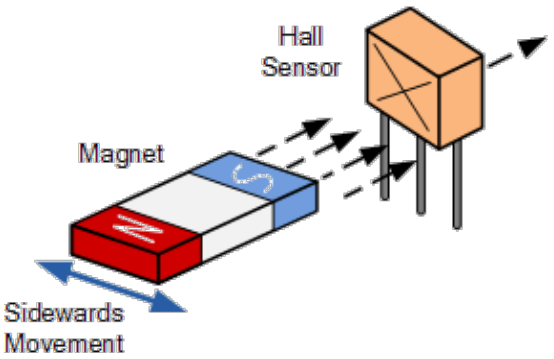
- It measures the position of the handle on the x-axis (right or left) and the y-axis (front/up or back/down)
- E.g., controlling speed and/or direction of mobile devices such as quadcopter and other robots

Hall effect sensor

- Hall effect sensors detect whether a magnet is **near** (no need to touch)
- 2 most common paths of motion for detecting a magnetic field:
  - **Head-on detection**

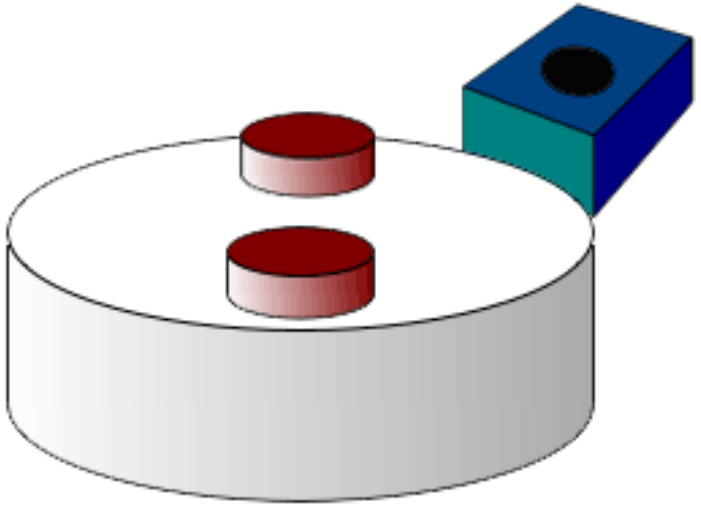


  - **Sideways detection**



Hall effect sensor: examples of application

- Hall effect sensors can be used to detect motion and to count motions without touching the thing you're detecting
  - Non-contact/waterproof type switches (e.g., detecting a door open/closed)
  - Rotary/shaft encoders (e.g., to count the number of revolutions of a wheel or to make a speedometer)



Sensors tracking biometric parameters

- Sensors that can track **biometric parameters** of a user
  - Heart beat
  - Galvanic skin response
  - Muscle tension
  - Electroencephalogram (EEG)
  - Breath
  - Gait (e.g., via shoes with embedded sensors)
  - Fingerprint (e.g., for authentication)

Sensors tracking properties of the environment

- Different types of environment properties can be tracked by dedicated sensors, e.g.:
  - Light sensors
  - Air pressure sensors
  - Pollution sensor
  - Humidity sensor
  - Microphones tracking audio signals (voice, music, environmental sounds)
  - Vibration sensors (for industrial machines, heartquake monitoring, etc.)

Actuators

- Actuators are devices that affect the world. They can provide different kinds of sensory stimuli, e.g.:
  - **Visual stimuli**
    - e.g., LED, screens, VR headsets, augmented reality apps
  - **Sound stimuli**
    - e.g., loudspeakers, headphones
  - **Haptic stimuli**
    - e.g., vibrotactile motors embedded in smartphones or wearables

- **Actuators can be used to provide feedback to a user in response to a certain status of the system**

Project: Sensors and actuators simulation

- In your project **you don't have to build a complete system with real sensors and actuators**, and there is no programming involved
- You can either augment an object with fake sensors or assume that an object already embeds those sensors
- You can simulate the sensors/actuators by attaching to a real object a paper/cardboard/other materials that indicate their presence onto/within the object
- The users who will try your system can **assume** that the object has those sensors that track a certain interaction
- If you know **AR tools** (e.g. Unity 3D), you can even simulate augmented reality-based interactions (using prerecorded videos with AR scenes to be embedded in Axure)

Project: Sensors and actuators simulation

- Instead, the app needs to be created in Axure and you can assume that your mobile device communicates **wirelessly** with the smart object or components of the smart environment
- Note: you can also assume that the app is running on a touchscreen directly embedded into the smart object
- **You need to design which are the gestures of users on your object and/or which quantities need to be measured and collected**
- How do you **repurpose** the quantities tracked by a sensor?

Types of interactions

- The interaction of a user with a system can be a combination of different types of interactions, e.g.:
  - Discrete or continuous
  - Touch-based or touchless
  - Synchronous or asynchronous
  - Individual or collaborative
  - Situated or remote
  - Networked
  - Unimodal or multimodal
  - …and others (tangible, embodied, ubiquitous, etc. …)
- **In designing your prototype you need to choose and reflect about the type of interaction you want to provide**

Discrete and continuous interactions

- **Discrete interactions**
  - The user interacts with the system in a discrete way: the user can manipulate a sensor that has a limited number of states, and get a discrete feedback in response to his/her actions
  - E.g., a user interacts with a push button
  - E.g., a beep sound is played in response to the clicked button
- **Continuous interactions**
  - The user interacts with the system in a continuous way: the user can manipulate continuously a sensor, and get a continuous feedback in response to his/her actions
  - E.g., a user interacts with a pressure sensor
  - E.g., a visual effect is displayed on screen which is controlled by the pressure sensor

Touch-based and touchless interactions

- **Touch-based interactions**
  - The user interacts with the system by physically manipulating an object
  - E.g., a user manipulates a joystick to control a drone

- **Touchless interactions**
  - The user does touch an object to make the interaction happen
  - E.g., infrared sensors or cameras track the gestures of a user in a videogame (e.g., Kinect)
  - E.g., hall effect sensor
  - E.g., QR code reader

Synchronous and asynchronous interactions

- **Synchronous interactions**
  - E.g., a user presses a finger onto a pressure sensors and receives immediately a sonic feedback as a result of this action

- **Asynchronous interactions**
  - E.g., a user pushes a button and the result of this action does not happen immediately, but it is triggered a different time

Individual and collaborative interactions

- **Individual interactions**
  - The user interacts alone with the system
  - E.g., a music streaming app for mobile phone

- **Collaborative interactions**
  - Multiple users interact together through the system
  - E.g., google doc, collaborative music making platforms, etc.

Situated and remote interactions

- **Situated interactions**
  - The user interacts with the system or other users, in the same physical space
  - E.g., a user playing a videogame, in the same room



- **Remote interactions**
  - The user interacts with the system or other users, in the different physical spaces (thanks to a communication network)
  - E.g., several users geographically displaced playing a game in VR



Networked interactions

- These are interactions between devices or between humans and devices, which are enabled by networks
- **Networked situated interactions**
  - E.g., a smartphone manipulates the parameters of a smart guitar
    - A musician controls in real-time via a tablet the smart guitar configuration while the guitarist is playing it (the smart guitar is a hub for collaborative music making)



Networked interactions

- **Networked remote interactions**
  - E.g., Telepresence systems (set of technologies that allow a person to feel as if others were present)
  - E.g., Industry 4.0: local operator using a machine and a remote operator communicating with him/her
  - E.g., Networked music performances
    - https://youtu.be/-rUOtor5-zI



Unimodal and multimodal interactions

- "**Mode**" refers to the communication through one sensory channel, to acquire or convey information

- **Unimodal interactions**
  - The user interacts with the system by using one single interaction modality as input or output
  - E.g., a user uses a mouse on a desktop computer, which provides a visual feedback

- **Multimodal interactions**
  - The user interacts with the system by using more than one interaction modality as input or output
  - E.g., virtual reality system with headset, haptic shoes, and tridimensional surround sound system

Multisensory interactive systems

- The term "multisensory" is used interchangeably with "multimodal"
- Multisensory systems use more than one mode for **input** and/or **output** of data, so the user is provided with multiple modes of interacting with a system
- **Multisensory systems aim at making communication with machines more natural, more efficient, and more engaging**

Multisensory vs Multimedia systems

- **Multisensory/multimodal** systems use more than one sense (or mode) of interaction
- **Multimedia** systems use a number of different media to communicate information
- E.g.,
  - A computer-based teaching system may use video, animation, text and still images: different media all using the visual mode of interaction
  - It may also use sounds, both speech and non-speech: two more media, now using a different mode

The human senses

- We perceive the world through the **combination** of our senses
- We have 5 senses + internal senses
- 5 senses:
  - **Taste**
  - **Touch**
  - **Audition**
  - **Olfaction**
  - **Vision**



- Internal senses:
  - **Vestibular system**: sense of balance
  - **Proprioceptive system**: sense for the position and movement of the body and limbs

User interfaces are evolving towards multisensoriality

- User interfaces are evolving from the most common unimodal GUI paradigm
- E.g., Amazon Echo Show employs voice as the primary input method but also includes a touchscreen



- Multisensory interfaces are an active area of research in HCI

Goals of multisensory interactive systems

- Two main goals:
  1. To achieve an interaction **closer to natural** human-human communication
  2. To increase the **robustness** of the interaction by using redundant or complementary information

- Multimodal products are more complex to design and implement than products that rely only on one modality (e.g., only visuals or only voice)

Advantages over unimodal systems

- Information coming from **multiple sources** can be
  - **Complementary**: bring additional information
  - **Redundant**: presence of overlapping information
- For both input (from the user) and output (to the user)
  - Multiple input sources may allow for better tracking
    - E.g., camera and microphone to track a spoken interaction
  - Multiple outputs: the user can be informed about an event through different sensory channels
    - E.g., interactive shoes that simulate a surface material both at auditory and haptic level simultaneously: more realistic

Advantages over unimodal systems

- Multimodal input
  - Provides users with multiple input modalities
  - Permits multiple styles of interaction
  - Allows humans to communicate more naturally with computers
- More input modalities can increase usability: the weaknesses of one modality are counterbalanced by the strengths of another
  - E.g., on a mobile device with a small visual interface and keypad, a word may be quite difficult to type but very easy to say (-> speech to text converter)

Advantages over unimodal systems

- More realistic, natural, engaging
- Flexible: different senses excel at different tasks
- Multimodal user interfaces have implications for accessibility: a well-designed multimodal application can be used by people with impairments
- The information that is presented via several modalities is merged and refers to **various aspects of the same process**

Questions when designing your smart object and its interactions

- What is the object you want to use or augment with sensors/actuators?
- Which are the sensors and actuators embedded in your object?
- What are the types of interactions for your project?
  - Discrete, continuous, or both
  - Touch-based, touchless, or both
  - Synchronous, asynchronous, or both
  - Individual, collaborative, or both
  - Situated, remote, or both
  - Networked
  - Unimodal or multimodal
  - …and the combination thereof

Homework

- Get together with your group and start discussing about the possible design of your smart object and its interactions
- Ground this discussion about the design space on the previous PACT and benchmarking phases

Bibliography

- Kranz, M., Holleis, P., & Schmidt, A. (2009). *Embedded interaction: Interacting with the internet of things*. IEEE internet computing, 14(2), 46-53.
- Robertson, G., Czerwinski, M., Fisher, D., & Lee, B. (2009). *Selected human factors issues in information visualization.* Reviews of human factors and ergonomics, 5(1), 41-81.